

calibration model, the volumes, and weights of the reagents, etc. Its general form is as follows:

General measurement model

$$Z = f(G_1, \dots, G_n, \dots, G_N) \quad (6.6)$$

Z represents the output quantity and $f()$ any mathematical relationship; it is the analyte concentration which will be inferred from the information collected by the N input quantities $G_1, \dots, G_n, \dots, G_N$ with $1 \leq n \leq N$. From a statistical point of view, the input quantities are the realizations of random variables with known probability distribution laws.

The example of a naive simple measurement model is given by the formula applied for calculating the concentration of the theophylline stock solution that was used to perform standard additions, as described in Section 5.2.2. To prepare such a solution, a known amount, m , of *analytical grade* theophylline is weighed on a laboratory scale. The purity P of this reagent is provided by the manufacturer and is, in this case, 99%. This mass is diluted in a known volume V of water. The measurement model which allows to obtain the concentration X_0 of the theophylline stock solution, is:

Measurement model for the stock solution

$$X_0 = \frac{m \times P}{V} \quad (6.7)$$

This measurement model contains three input quantities, i.e. $N = 3$. It allows us to establish clearly and without ambiguity what is going to be measured and represents the quantitative expression which links the value of the measurand to the three input quantities on which it depends.

At this stage, it is interesting to verify the consistency of the units. Thus, X_0 will evidently be expressed as mg/ml because m is measured in mg, V in ml, and quantity P is dimensionless since it is a purity level. Later, for sample spiking, this stock solution is diluted and introduced such as the taken volume V_p and the final dilution volume V_f gives the expected final concentration as shown in this new model:

$$X_1 = \frac{X_0 \times V_p}{V_f}$$

But measurement models are often more complex and may combine many input quantities that are themselves other measurands. For example, the determination of lead by an ICP emission spectrophotometry with isotope dilution and coupling with a mass spectrometer or inductively coupled plasma-isotopic dilution-mass spectrometry (ICP-ID-MS) is an illustration of a complex measurement model [7]. In this case the model includes $N = 12$ input quantities:

Measurement model

$$Z = \left(\frac{Ms - Cs}{Pp.Wa} \right) \times \left(\frac{K.As6 - As8}{Ap8 - K.Ap6} \right) \quad (6.8)$$

Correction coefficient

$$K = \frac{\left(\frac{Ar8}{Ar6} \right)}{Rr} \times Rp \quad (6.9)$$

Table 6.1 contains the list of all input quantities with their definitions and units. It also provides an observed value G_n for each that is used to compute an example lead concentration measurement value on a future CRM. It is important to keep all significant digits to make such a calculation. The observed measurement value is 1.999982505 mg/kg.

For this type of complex model, a preliminary precaution consists in checking what is called the dimensional equation. An error in the mathematical formulation can thus be corrected. In the case of Eq. (6.8), it is easy to check that the result of the dimensional equation is mg/kg, with most quantities being dimensionless ratios.

This approach to MU, based on the measurement model, is favoured by metrologists. However, the restriction to such measurement model is incomplete when classical analytical operations have to be considered, such as sampling in medical biology or sample preparation. Chapter 7 describes a more comprehensive method which we consider as better suited to analytical sciences and derived from the method accuracy profile (MAP).

Table 6.1 LEAD – input quantities for the lead measurement model by ICP-ID-MS (CRM: certified reference material).

Description of the input quantities	Symbol	Unit	G_n
Mass of the spike	M_s	g	0.7806
Pb concentration in the spike	C_s	mg/kg	0.4149
Mass of the test portion	M_p	g	0.4944
Moisture (water activity)	W_a	%	0.9255
Isotopic abundance of ^{206}Pb in the CRM	Ar_6	%	40.0890
Isotopic abundance of ^{208}Pb in the CRM	Ar_8	%	40.0954
Ratio $^{208}\text{Pb}/^{206}\text{Pb}$ in the CRM	R_r	%	1.0189
Ratio $^{208}\text{Pb}/^{206}\text{Pb}$ in the test portion	R_p	%	0.8994
Isotopic abundance ^{206}Pb in the spike	As_6	%	0.9997
Isotopic abundance ^{208}Pb in the spike	As_8	%	0.0001
Isotopic abundance ^{206}Pb in the test portion	Ap_6	%	0.2454
Isotopic abundance ^{208}Pb in the test portion	Ap_8	%	0.5290
Output quantity			
Pb concentration in the test sample	Z	mg/kg	1.999982505

There are also operating procedures where the analyte is defined by the method itself. This type of method is referred to as the “criteria method” by the Codex Alimentarius Commission [8]. The analyte can also be said to be defined *per se*, such as the determination of fiber or moisture in foods. Most often, the ultimate step consists of a series of weighings. They are gravimetric methods, and the uncertainty of the scales is used to calculate their combined MU.

It is frequent for traditional and long-standing standardized methods defined *per se* to be used as a reference to validate a new alternative method. This is the case when the so-called rapid microbiological counting method is used in food microbiology as an alternative to substitute for a traditional method using Petri dishes. The Association of Official Analytical Chemists (AOAC) has proposed to call this type of traditional reference method a *gold standard*. A common surprise is that the MU of the reference method may be higher than that of the alternative method. An example of a comparison of an alternative method to a reference method using the MAP and MU is presented in Section 10.2.2.

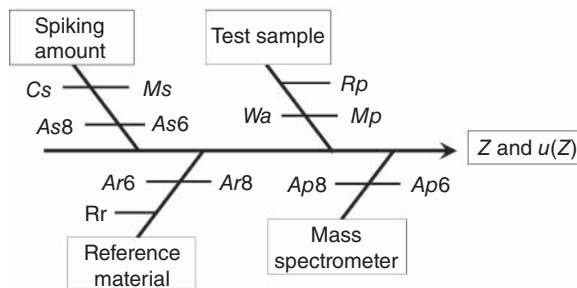
6.5 Stage 2. Identify Uncertainty Components

The second phase of the GUM procedure is crucial because it determines the rest of the process. From a practical point of view, it is advisable to use a cause-effect diagram or fishbone diagram to list the components (or sources) of the MU, identify their origin, and predict their influence on the overall uncertainty. It also avoids counting the same source multiple times and provides a simple mnemonic framework. This practical (and recommended) means is illustrated in Figure 6.2 for the ICP-ID-MS method of lead determination. All input quantities and the output quantities described in Table 6.1 appear in this diagram.

Each edge corresponds to an input quantity, considered as a potential source of uncertainty. It would be possible to go into more details. For example, all quantities involved in the preparation of Pb stock solution used for spikes at C_s concentration could be added, such as volume, weights and lead purity corresponding to the model (6.7). Similarly, double weighing is used to quantify the moisture contents W_a .

However, in view of the considerable number of operations and quantities that lead to a chemical or biochemical measurement value, it is necessary to limit it to a

Figure 6.2 LEAD – cause and effect diagram of the sources of uncertainty when determining lead by ICP-ID-MS.



reasonable level of detail. It is therefore a rather delicate stage, the main aim of which is to provide a perfectly clear and operational vision of what needs to be considered later.

The main problem with this approach is that the analyst has limited the measurement model to the reagents involved. Many other sources of uncertainty should be considered, such as extraction time, temperature, the aging of the reagents, and the skill level of the personnel. All these quantities are well known in the analytical science to have a definite influence on the result.

But the sources of variability, if they were to be included in the model in the same way as a volume or a weight, would pose serious modeling problems. At this stage, it seems difficult to put the various operations of mineralization, dry-ashing, or extraction into a mathematical equation! As claimed, in Chapter 7 a more empirical, comprehensive approach, sometimes called holistic, is proposed that is adapted to the practice of analysts and makes MU estimation simpler.

6.6 Stage 3. Quantify Uncertainty Sources

The third stage includes the following operations. For each previously identified component, it is required to quantify its amplitude of variation. In statistical terms, this means obtaining an estimate of the standard deviation that characterizes each random variable associated to an input quantity. However, if it seems impractical or cumbersome to quantify them experimentally one by one, the GUM recommends grouping the sources of uncertainty together, “as far as possible.”

For example, if one identifies, on the one hand, variations in the setting of the measuring instrument, and, on the other hand, the aging of the reagents, as components to be evaluated, they can be grouped together by making measurements on different days, modifying the instrumental settings and the reagents from one day to another. GUM defines two ways to quantify component variability. This point is fundamental.

Type A evaluation	It is empirical since it is based on statistical analysis of series of observations; some documents call it the <i>top-down</i> approach.
Type B evaluation	It is more deterministic, since it does not use observations but theoretical probability laws, chosen a priori; it is also called the <i>bottom-up</i> approach.

From a practical point of view, and whatever the method of evaluation of each source (or grouping of sources), the aim is to obtain a statistical parameter of dispersion. Under pressure from statisticians, metrologists have given up calling the standard deviation this estimate of dispersion. Especially since, as explained in the next chapter, these parameters can be recombined, sometimes in a rather unconventional way. The term *standard uncertainty* instead of standard deviation has thus come into use.

By convention, the standard uncertainty of the input quantity G is denoted $u(G)$. First, for each input quantity, it is essential to choose the most appropriate method

of evaluation of the standard uncertainty, either type A or type B. When several quantities are included in a measurement model, the combination of both types of evaluation is completely appropriate.

For example, in the case of lead determination by ICP-ID-MS, both modes of standard uncertainty evaluation were used. Thus, the $u(Ms)$ was estimated by a type B approach based on the probability law provided by the balance manufacturer, while $u(Wa)$ is the standard deviation of a series of repeated moisture measurements.

Next, at the fourth stage, the GUM describes how to combine the individual standard uncertainties of each source into an overall standard uncertainty called the combined standard uncertainty denoted $u_c(Z)$. In the rest of this text, the terms standard uncertainty and standard deviation may be indifferently used, with standard variance when it is a squared standard uncertainty.

6.6.1 Type A Approach

This approach starts by collecting observed measurement values purposefully obtained or from archive. The data that can be used to estimate a standard uncertainty are of various origins, and the following proposition list is not exhaustive.

6.6.1.1 Accuracy Profile

We consider that MAP is probably the most suitable data source for an initial estimate of MU. Section 7.2 is entirely devoted to the details of the calculations from this type of data. In this case, the combined standard uncertainty can directly be obtained from the measurements made on the diverse validation materials.

An important constraint is that measurements must be collected according to a properly structured experimental design and not randomly. The rules to correctly organize the experimental design for collecting the suitable data to construct an accuracy profile are thoroughly explained in Section 5.4.2. The purpose of this organization is to trigger as many sources of uncertainty as possible and run the trials in separate series under intermediate precision conditions.

For example, if the replicates are scattered over several days, it is then possible to change the operators, the settings of the measuring system, the reagents, the calibration curve, etc. This strategy is only possible if the test material can be conserved long enough without undergoing a change considered too important, or prepared at will, for instance by weighing. If not, it will be necessary to resort to another organization of the replicates, which makes it possible to vary the sources of uncertainty that normally occur in the framework of routine usage of the method.

6.6.1.2 Interlaboratory Study

If the laboratory takes part in an interlaboratory analysis, the collected measurements allow to estimate the MU of the test sample analyzed by all participants. It is recommended that the grand mean, estimated at the end of the study, be used as a reference value. The situation is even better if the analyzed material is a CRM. However, the standard deviation of reproducibility s_R cannot be directly used